

torch.argsort#

<https://pytorch.org/docs/stable/generated/torch.argsort.html>

Description:

Returns the indices that sort a tensor along a given dimension in ascending order by value. This is the second value returned by `torch.sort()`. See its documentation for the exact semantics of this method. If `stable` is `True` then the sorting routine becomes stable, preserving the order of equivalent elements. If `False`, the relative order of values which compare equal is not guaranteed. `True` is slower. `input` (Tensor) – the input tensor. `dim` (int, optional) – the dimension to sort along

Function Signature

```
torch.argsort(input, dim=-1, descending=False, *,  
stable=False) → Tensor#
```

Parameters

Keyword Arguments

Parameters

Keyword

stable (bool, optional) – controls the relative order of equivalent elements

Code Examples

```
>>> a = torch.randn(4, 4)
>>> a
tensor([[ 0.0785,  1.5267, -0.8521,  0.4065],
        [ 0.1598,  0.0788, -0.0745, -1.2700],
        [ 1.2208,  1.0722, -0.7064,  1.2564],
        [ 0.0669, -0.2318, -0.8229, -0.9280]])

>>> torch.argsort(a, dim=1)
tensor([[2, 0, 3, 1],
        [3, 2, 1, 0],
        [2, 1, 0, 3],
        [3, 2, 1, 0]])
```

Detailed Documentation

Documentation

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This is the second value returned by `torch.sort()`. See its documentation for the exact semantics of this method.

If `stable` is `True` then the sorting routine becomes stable, preserving the order of equivalent elements. If `False`, the relative order of values which compare equal is not guaranteed. `True` is slower.

`input (Tensor)` – the input tensor.

`dim (int, optional)` – the dimension to sort along

`descending (bool, optional)` – controls the sorting order (ascending or descending)

`stable (bool, optional)` – controls the relative order of equivalent elements